

Table 9-2: *Python-Only Turtle Functions*

Python	Description
<code>begin_fill()</code>	Begins drawing a filled-in shape. The lines drawn after this call will specify the perimeter of the filled-in shape.
<code>end_fill()</code>	Draws the filled-in shape that was started with the call to <code>turtle.begin_fill()</code> .
<code>fillcolor(<i>color</i>)</code>	Sets the color used for filled-in shapes.
<code>hideturtle()</code>	Hides the triangle that represents the turtle.
<code>showturtle()</code>	Shows the triangle that represents the turtle.
<code>tracer(<i>drawingUpdates</i>, <i>delay</i>)</code>	Adjusts the speed of drawing. Pass 0 for <i>delay</i> for a delay of 0 milliseconds after each line the turtle draws. The larger the number passed for <i>drawingUpdates</i> , the faster the turtle draws by increasing the number of drawings before the module updates the screen.
<code>update()</code>	Draws any buffered lines (explained later in this section) to the screen. Call this after the turtle has completed drawing.
<code>setworldcoordinates(<i>llx</i>, <i>lly</i>, <i>urx</i>, <i>ury</i>)</code>	Readjusts which part of the coordinate plane the window shows. The first two arguments are the x, y coordinates for the lower-left corner of the window. The latter two arguments are the x, y coordinates for the upper-right corner of the window.
<code>exitonclick()</code>	Pauses the program and closes the window when the user clicks anywhere. Without this at the end of your program, the turtle graphics window may close as soon as the program ends.

In Python's `turtle` module, lines are displayed on the screen immediately. However, this can slow programs that draw thousands of lines. It's faster to *buffer*—that is, hold off displaying several lines and then display them all at once.

By calling `turtle.tracer(1000, 0)`, you can instruct the `turtle` module to hold off displaying lines until 1,000 lines have been created by your program. After your program has finished calling line-drawing functions, make a final call to `turtle.update()` to display any remaining buffered lines to the screen. If your program is still taking too long to draw an image, pass a larger integer such as 2000 or 10000 as the first argument to `turtle.tracer()`.